

[illegible]

The diagram shows the pin connections for an ESP32-53-WROOM-1U-N16R2 module. The module is connected to a 3V3 power source. The TX pin (pin 3) is connected to TXD0 (pin 37). The RX pin (pin 2) is connected to RXD0 (pin 36). The LED_DATA pin (pin 16) is connected to TXD0 (pin 37). The TX pin (pin 3) is also connected to TXD0 (pin 37). The RX pin (pin 2) is also connected to RXD0 (pin 36). The LED_DATA pin (pin 16) is also connected to TXD0 (pin 37).

Module Pin	Module Label	Module Pin	Module Label
3	EN	37	TXD0
2		36	RXD0
27	IO0	35	IO1
28	IO1	34	IO2
39	IO2	33	IO3
38	IO3	32	IO4
35	IO4	31	IO5
34	IO5	30	IO6
33	IO6	29	IO7
32	IO7	28	IO8
31	IO8	27	IO9
30	IO9	26	IO10
29	IO10	25	IO11
28	IO11	24	IO12
27	IO12	23	IO13
26	IO13	22	IO14
25	IO14	21	IO15
24	IO15	20	IO16
23	IO16	19	IO17
22	IO17	18	IO18
21	IO18	17	IO19
20	IO19	16	IO20
19	IO20	15	IO21
18	IO21	14	IO22
17	IO22	13	IO23
16	IO23	12	IO24
15	IO24	11	IO25
14	IO25	10	IO26
13	IO26	9	IO27
12	IO27	8	IO28
11	IO28	7	IO29
10	IO29	6	IO30
9	IO30	5	IO31
8	IO31	4	IO32
7	IO32	3	IO33
6	IO33	2	IO34
5	IO34	1	IO35
4	IO35	0	IO36
3	IO36	35	IO37
2	IO37	34	IO38
1	IO38	33	IO39
0	IO39	32	IO40
		31	IO41
		30	IO42
		29	IO43
		28	IO44
		27	IO45
		26	IO46
		25	IO47
		24	IO48
		23	IO49
		22	IO50
		21	IO51
		20	IO52
		19	IO53
		18	IO54
		17	IO55
		16	IO56
		15	IO57
		14	IO58
		13	IO59
		12	IO60
		11	IO61
		10	IO62
		9	IO63
		8	IO64
		7	IO65
		6	IO66
		5	IO67
		4	IO68
		3	IO69
		2	IO70
		1	IO71
		0	IO72

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Pinout diagram for the ATmega328P microcontroller. The diagram shows 18 pins with their functions and power supply connections. Pins 1 and 2 are labeled TP1 and TP2, connected to +3V3 and +5V respectively. Pin 3 is TP3, connected to GND. Pins 4 through 11 are labeled TP4 through TP11, connected to SPI_MOSI, SPI_SCLK, SPI_MISO, I2C_SDA, I2C_SCL, TWI_CTRL, TW2_CTRL, and CTA_P respectively.

SSR terminals 14/9J: $j_{p1} = SSR_{EN}(\text{board} + \text{v5v, watchdog-gated})$
 $j_{p2} = \text{the switched low side, Board-safe: } H720/\text{hold each} \\ \text{input gate down while the E5372 pins are high impedance.}$
 NOT optional: $j_{p3} = \text{level 1 and level 2 generated: it an opto}$
 only isolates if the SSR loop is powered off-and-on; an opto
 that powers it.

TC1 terminal J3 / TC2 terminal J8: $j_{p1} = K_{+}$, $j_{p2} = K_{-}$ – both
 tied, biased near AGND only through each MAX485's internal
 BIOS network. Ungrounded-junction boards required: two
 grounded-junction boards in one kin would loop through it.

Nav switch J6 is panel-mounted; Inputs use ESP32 pull-ups.

J11: $IN1/IN2 \rightarrow IN3$ (lilo/gas-flow/spare) + GND – dry contact each
 channel to GND.

J13: DNP SELV voltage sense header. NOT mains-rated, not fitted
- do not wire to AC mains. Y1/C26 load caps are an ASSUM
value (unverified C.L), see design.py comment. ADE7953 I2C
address 0x38 collides with PCF8574A.